Minimax Optimality of SCX Noise Detection — Feasibility Analysis

Purpose: Assess whether a matching lower bound for Theorem 1's exponential rate can be proved, establishing minimax optimality.

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1. Executive Summary

Verdict: MEDIUM feasibility. A minimax lower bound establishing that the exponent `exp(-cMΔ²)` is optimal (up to constants) is achievable, but **only for a simplified version of the problem**. Proving the exact constant `c = 2` (matching Hoeffding) under the full SCX model with F1 risk is significantly harder and may require substantial mathematical development.

Two separate minimax problems must be distinguished:

Regime	Parameter	Upper bound	Lower bound difficulty	Status
M-regime (experts)	M = #experts	`exp(-2MΔ²)`	**Hard**	No lower bound exists

Regime	Parameter	Upper bound	Lower bound difficulty	Status
- regime (features)	= I(; S)	`O(√ )`	**Standard**	Trivially matches Thm 2

The JMLR PAPER_FRAMEWORK correctly identifies the need for a matching lower bound but significantly underestimates the difficulty of the M-regime (listing "4 days" for Le Cam + Assouad). The -regime bound is indeed a 4-day project. The M-regime bound is more realistically **4-8 weeks** of focused work by a trained mathematical statistician.

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2. Feasibility Assessment by Regime

2.1 M-Regime: Exponent in Number of Experts — HARD

Goal: Show that for any noise detector observing M experts, `inf_ sup_P (1 - F1( , P))  C • exp(-cMΔ²)` for some `c  2`.

What makes it hard (3 distinct obstacles):

1. **Expert dependence under noise** (the core technical obstacle):
    - Under CLEAN data, experts are conditionally independent given x (Assumption A2).
    - Under NOISY data, experts SHARE the noise label y y*. Given the noise label c, they're conditionally independent; but marginally over c, they become dependent.

- The current Theorem 1 proof handles this by conditioning on c and applying a union bound. For the LOWER bound, we can't do this — the marginal dependence structure is fundamental.
- **Implication**: Standard lower bound techniques (Le Cam, Assouad) are designed for product distributions. The mixture structure under noise complicates divergence calculations.

2. **F1 is a non-standard risk for minimax analysis**:
   - F1 involves both false positives and false negatives in a non-linear way.
   - Standard minimax theory (Lehmann & Romano, Tsybakov) focuses on 0-1 classification risk, squared error, or Lp losses.
   - Converting F1 lower bounds to 0-1 error lower bounds requires care: `1 - F1 = (FP + FN) / (2TP + FP + FN)` which depends on the base rate .
   - Most minimax toolkits assume a risk function that decomposes sample-wise; F1 does not (it's a population-level summary).
3. **State conditioning and threshold selection**:
   - The lower bound must account for the optimal threshold * and state partition Π.
   - A hard instance must simultaneously be "hard for any detector" while respecting the state structure.
   - This requires constructing distributions where the separation gap Δ is exactly the parameter of interest, and the detector cannot reliably distinguish clean from noisy samples.

2.2 -Regime: Feature Weakness — STANDARD (well-understood)

Goal: Show that `inf_  sup_{P: I( ;S)   } (F1_base - F1( , P))   C • √ .`

Why it's standard: - This is a classic information-constrained testing problem. - Le Cam's two-point method with TV perturbation works directly. - Theorem 2 of Paper 1 already proves `F1_SCX   F1_base + C_F·√( /2)`. - A matching lower bound just needs the reverse inequality: `F1_SCX   F1_base - C·√ ` (i.e., SCX cannot be outperformed by any method). - This is essentially already covered by the "minimax optimality" claim in Theorem 2's symmetric lower bound (Eq. 377 of the current file): `AUC(h_SCX)   AUC_base - √( /2)·(1/  + 1/(1- ))`. - Actually, the JMLR paper's minimax claim for Section 3.2 is about the -regime, and this IS already largely done.

Caveat: The existing "minimax" claim in Theorem 2 is only a LOWER bound for SCX (showing it can't beat baseline when features are weak). The true minimax claim would be the CONVERSE: no OTHER detector can beat SCX's rate. This inverse direction requires showing SCX is optimal among all methods, not just that it degrades gracefully.

2.3 The Gap in the JMLR Plan

The JMLR PAPER_FRAMEWORK conflates these two regimes. Section 3.2 is titled "Weak Feature Lower Bound" (-regime) but claims to prove a minimax lower bound matching Theorem 1's upper bound (M-regime). The section then says:

> "The gap between `O(exp(-2MΔ²))` and `O(√ )` is not a contradiction: the upper bound is for expert variance M, while the lower bound is for feature information ."

This is conceptually confused. A truly matching lower bound for the M-regime would show `Ω(exp(-cMΔ²))` — this is what is missing, and it is NOT what the current Section 3.2 delivers.

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3. Technical Approach: What Would Work

3.1 Preferred Method: Le Cam's Two-Point Method (M-regime)

Setup: Construct two data-generating processes P and P that: - Are hard to distinguish (bounded divergence) - Have different noise indicators for a specific sample

Construction (sketch):

Fix a single state s with K classes. Consider M experts trained on disjoint data.

- **P (clean)**: For sample (x, y*), each expert makes error with probability   (independently, given x). The consistency score C(x) ~ (1/M) • Binomial(M,).
- **P (noisy)**: Sample has noise label y   y*. Given noise label c, each expert makes error with probability 1 - _c, where _c = P(f_m(x) = c | x). With A6 (balanced errors), 1 - _c   1 - /(K-1).

The key divergence calculation: `TV(P ^M, P ^M)` where P ^M is the joint distribution of (e ,…,e_M) under clean, and P ^M is the joint under noise.

For product distributions (experts conditionally independent given the noise label):

Under **clean** (P): `(e ,...,e_M) ~  _m Bernoulli( )` — product distribution.

Under **noise** (P): `(e ,...,e_M) ~ (1/(K-1))• _{c y*}  _m Bernoulli(1- _c(x))`

— mixture of products.

For the lower bound, we upper-bound TV(P , P) and apply Le Cam's lemma:

`inf_  max{P ( =1), P ( =0)}   (1 - TV(P ^M, P ^M))/2`

The ² divergence approach (most promising):

For a single expert: `²(Bern( ) || Bern(1- /(K-1))) = (1 -  /(K-1) - )² / ((1- /(K-1))•( /(K-1)))`

For M experts under clean (product) vs noise (mixture), we use the fact that TV √(²/2) and:

`²( Bern( ) || (1/(K-1)) Bern(1- _c)) = ?`

The mixture denominator makes this non-trivial. One can use Jensen to bound:

`²(P  || P )   (1/(K-1))• _c  ²(P  || P ,c)`

where P ,c is the conditional distribution under noise label = c. Then:

`²(P  || P ,c) = [(1- /(K-1) -  )² / ( (1- /(K-1))•( /(K-1)) )]^M`

This gives `TV(P , P )   √( ²/2)   √( (1/(K-1))•[1 + O(MΔ²)]^M ) /` √2 which is too loose for large M.

The M-dependence in the exponent is the core difficulty.

3.2 Alternative: Fano's Method with Packing

Why not Fano: Fano's inequality is typically used for MULTI-way hypothesis testing (K 3 hypotheses). For a TWO-point problem (clean vs. noise), Fano reduces to Le Cam. The advantage of Fano is for problems with many states (K states → 2ˆK hypercube for Assouad). But the M-regime is fundamentally a two-point problem (test noise vs. clean), so Fano doesn't add value here.

3.3 Alternative: Assouad's Method

When Assouad helps: For -regime with K states, Assouad gives the `Ω(√K • √ )` rate. This matches the `|S|` factor in the upper bound.

For M-regime: Assouad doesn't naturally apply because the parameter of interest (noise detection rate) is not a hypercube of binary parameters.

3.4 Recommended Strategy: Three-Step Proof

Step 1 (Easier): Prove a LOWER BOUND ON TESTING ERROR, not F1. - Show that for any detector : P(correctly classifies sample as clean|noise vs clean)   C • exp(-cMΔ^2) - Use Le Cam with carefully constructed P (all experts make errors at rate) and P (experts make errors at rate 1- /(K-1)) - Key lemma: TV(Bin(M,) || Bin(M, 1- /(K-1)))   √(²) bound - This gives: inf_ max{P (mistake), P (mistake)}   (1/4) • exp(-2MΔ^2) for reasonable parameters

Step 2 (Harder): Convert testing error to F1 lower bound. - Show that `1 - F1( )   ( /2) • P(error on noisy samples)` - This requires bounding the false positive rate contribution - The conversion loses constant factors but preserves the exponential rate

Step 3 (Currently speculating): Show that the Hoeffding exponent `c = 2` is tight. - This requires a Berry-Esseen type refinement: the CLT approximation to the binomial tail shows that exp(-2MΔ^2) is the exact Gaussian approximation - For exact tightness: show lim_{M→∞} (-1/M) log(inf_ max-error) = $2\Delta^2$ - This is a large deviations rate result, provable via the Cramer-Chernoff theorem for i.i.d. Bernoulli variables

Critical observation: The large-deviations rate function for Bernoulli(p) is KL(p||q). The Hoeffding bound gives exp(-2(p-q)^2), while the exact large-deviations rate is exp(-M • KL(p||q)). By Taylor expansion, KL(p||q) = 2(p-q)^2 + O((p-q)^3). So: - For SMALL gaps $\Delta \to 0$: Hoeffding is tight (both $2\Delta^2$) - For LARGE gaps $\Delta \to 1/2$: Chernoff gives significantly tighter exponent (KL > $2\Delta^2$) - The exact minimax rate is `exp(-M•KL( || ) + o(M))` not `exp(-2MΔ²)`

This means: the **rate** (exponential decay in M) is optimal, but the **constant** 2 in the exponent can be improved by using the Chernoff form. The optimal constant depends on the specific Bernoulli parameters.

3.5 Summary of What is Provable

Statement	Difficulty	Method	Timeline
`inf_ sup_P (Bayes error) C•exp(-cMΔ²)`	Medium	Le Cam + ^{2}	2-3 weeks
`liminf(-1/M) log(inf_ sup_P (Bayes error)) = 2Δ²`	Medium-Hard	Cramer-Chernoff + CLT	3-4 weeks

Statement	Difficulty	Method	Timeline
Same for F1 risk	Hard	Requires converting testing error to F1	+2-3 weeks
Exact constant c = 2 for all finite M	Very Hard	May require new concentration tools	>3 months
-regime minimax (matching Thm 2)	Standard	Le Cam (already essentially done)	<1 week

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4. What the Result Would Look Like

4.1 Conjectured Theorem (M-Regime)

Theorem (Minimax Lower Bound for Consistency-Based Noise Detection). Assume B1-B3 (conditional independence, state-conditional noise, non-degeneracy). For any state s with separation gap $\Delta > 0$, M 2 experts, noise rate (0, 1/2), and class count K 2:

Part (a) — Testing lower bound:

$$\inf_{\psi} \sup_{P \in \mathcal{P}_\Delta} [\mathbb{P}_P(\psi = 1 \mid \text{clean}, s) + \mathbb{P}_P(\psi = 0 \mid \text{noise}, s)] \geq \frac{1}{2} \exp(-2M\Delta^2)$$

where   is any measurable noise detector, and P_Δ is the set of distributions satisfying the separation condition with gap   Δ.

Part (b) — F1 lower bound:

$$\inf_{\psi} \sup_{P \in \mathcal{P}_\Delta} [1 - \mathrm{F1}(\psi, P)] \geq \frac{\eta_{\min}}{4} \cdot \exp(-2M\Delta^2)$$

where _min = min(, 1-).

Part (c) — Rate optimality: The SCX consistency detector achieves the exponent 2MΔ² in its F1 guarantee (Theorem 1), and no detector can achieve exponent > 2MΔ² (by Part (b)). Therefore, SCX is **minimax rate-optimal**.

Corollary (Large-Deviations Refinement):

$$\lim_{M \to \infty} -\frac{1}{M} \log \inf_{\psi} \sup_{P \in \mathcal{P}_\Delta} [1 - \mathrm{F1}(\psi, P)] = 2\Delta^2$$

confirming that the exponent 2Δ² per expert is the information-theoretic limit.

4.2 Conjectured Theorem (-Regime — matching Thm 2)

Theorem (Minimax Lower Bound for Weak Feature Noise Detection). Let   be a feature mapping with I((X); S)   . For any noise detector h operating on ((X), Y, {f_m(X)}):

$$\inf_{h} \sup_{P: I(\phi; S) \leq \delta} (\mathrm{F1}(h) - \mathrm{F1}_{\text{base}}) \geq 0$$

and

$$\sup_h \inf_{P: I(\phi;S)\leq\delta} (\mathrm{F1}(h) - \mathrm{F1}_{\mathrm{base}}) \leq C_F \cdot \sqrt{\frac{\delta}{2}}$$

i.e., SCX achieves the minimax-optimal rate O(√) in the weak-feature regime.

(Note: This is essentially already Theorem 2's symmetric bound. The only missing piece is the "inf over h" part — proving no method can beat SCX's rate.)

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5. Key Challenges (Detailed)

Challenge 1: Marginal Dependence Under Noise (MOST SERIOUS)

The problem: - Under noise, P(e ,…,e_M | noise) = (1/(K-1)) • _c _m P(e_m | noise, c) - This is a MIXTURE of product distributions, NOT a product distribution - Standard minimax techniques (Le Cam, Assouad) rely on divergence tensorization for product distributions - The mixture structure breaks tensorization

Possible resolution: Use the fact that: `TV(P , P )  (1/(K-1)) • _c TV(P , P ,c)` by the convexity of TV in its second argument.

Then TV(P , P ,c) √(²(P || P ,c)/2) where P ,c is product.

This gives: `TV(P , P )  √( ²(P || P ,c)/2)` for the worst-case c.

Remaining gap: This bound might not be tight for moderate M. The mixture structure could make the two distributions EASIER to distinguish, meaning the bound is loose (lower bound becomes too weak). This is acceptable — a weak lower bound is still a valid lower bound for minimax purposes — but it limits the sharpness of the result.

Challenge 2: F1 Risk Does Not Factorize

The problem: - F1 is a ratio of expectations, not an expectation of a ratio - Most minimax lower bound techniques work for additive risks (0-1 error, MSE) - F1 involves TP, FP, FN jointly through: F1 = 2TP / (2TP + FP + FN)

Possible resolution: - Bound 1 - F1 from below by the testing error: `1 - F1  •(1 - TPR) + (1- )•FPR` (using the inequality in Theorem 1's proof) - Then lower bound on testing error (Challenge 1) directly gives a lower bound on 1 - F1 - This loses a factor of but preserves the exponential rate

Remaining gap: The inequality `1 - F1  •FN_rate + (1- )•FP_rate` is not tight. The true F1 lower bound has denominator effects. This is acceptable for rate optimality but loses constant factors.

Challenge 3: Separation Gap Δ Must Be "Known" to the Detector

The problem: - A minimax lower bound must hold for ANY detector, even one that knows Δ exactly - The lower bound construction should NOT give the detector any advantage from knowing Δ - Actual SCX uses a fixed threshold ; the minimax optimal detector might use a different threshold

Resolution: Construct P and P such that the optimal Bayes classifier is exactly `  = 1{C(x) >  }` for some . Then any detector must pay the minimax price. This is feasible by making the clean and noisy distributions have different means.

Challenge 4: Proving the Exact Constant c = 2

The problem: - For finite M, the Hoeffding constant 2 is not necessarily tight - The exact minimax rate depends on Bernoulli parameters via: `lower_bound  exp(-M • inf_{t  [0,1]} KL(t || separating threshold))` - For Bernoulli( ) vs Bernoulli(1- /(K-1)), the optimal threshold is at * = (+ 1- /(K-1))/2 - The KL divergence KL(* ||) = 2(- *)² + O((-*)³) by Taylor expansion - So for Δ → 0: constant = 2 exactly (tight) - For Δ > 0.1: constant > 2, meaning Chernoff lower bound gives BETTER rate than exp(-2MΔ²)

Bottom line: The exponent `2MΔ²` is rate-optimal but not constant-optimal. A more precise lower bound would be: `exp(-M • KL( * ||  ))` where * = (+ 1- /(K-1))/2

This is larger than exp(-2MΔ²) for Δ > 0.

Challenge 5: State Aggregation

The problem: - Theorem 1's F1 bound aggregates over states s via _s - A minimax lower bound should also allow state aggregation - The presence of multiple states with different Δ_s values complicates the construction

Resolution: Focus on the hardest state s* = argmin_s Δ_s. Construct the lower bound for this single state. The multi-state extension follows by setting _s* = 1 (all probability mass on the worst state). This is a valid hard instance within the problem class, giving the correct minimax rate.

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6. Value to JMLR Paper

6.1 Critical Assessment

Aspect	With Lower Bound	Without Lower Bound
Novelty of Thm 1	Complete statistical theory	Upper bound only (one-sided)
Claim of "optimality"	**Can claim SCX is minimax optimal**	Cannot claim optimality
Relation to Thm 2	Symmetric: upper bound AND lower bound	Only lower bound for weak features
JMLR acceptance	**Strong differentiator**	Still strong, but one gap noted by reviewers
Reviewers' likely reaction	"Comprehensive theory, well-rounded"	"Where is the matching lower bound?"

Verdict: A matching lower bound would elevate Paper 3 from "good theory paper" (acceptance likely) to "exceptional theory paper" (highly cited, talked about). Reviewers at JMLR level will notice the asymmetry between Theorem 1 (upper bound only) and Theorem 2 (upper+bound + lower bound). They may ask: "Why do you provide a lower bound for feature weakness but not for the statistical rate?"

6.2 What JMLR Reviewers Will Ask

1. "Is the exponent $2M\Delta^2$ tight? Could there be a different method that achieves $3M\Delta^2$?"
    - Answer without lower bound: "We conjecture it's tight based on Hoeffding"
    - Answer with lower bound: "No, the minimax rate is at most $2\Delta^2$ per expert"
2. "The paper claims SCX is optimal — optimal compared to what?"
    - Answer without lower bound: "Optimal among consistency-based detectors" or silence
    - Answer with lower bound: "Minimax optimal among ALL detectors operating on the same data"
3. "How does Theorem 1 relate to the minimax claim in Section 3.2?"
    - Answer without a proper M-regime lower bound: Confusing (different regimes)
    - Answer with proper bounds: "SCX is rate-optimal in both M and regimes"

6.3 Reviewer Risk

Without a matching M-regime lower bound, a knowledgeable reviewer (e.g., someone who knows minimax theory from either Tsybakov's or Wainwright's textbook) will likely:

- **Ask for it**: "The authors should provide a matching lower bound or discuss why the upper bound is tight"
- **Not reject solely for this**: JMLR papers don't always need matching lower bounds, especially for novel frameworks

- **Downgrade novelty claim**: From "optimal" to "near-optimal" or "exponential"

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7. Estimated Effort

7.1 Pessimistic Estimate (Academic Mathematician/Statistician)

Component	Effort	Description
Le Cam construction for Bernoulli testing	1-2 weeks	Standard, but needs careful calculation of ^{2} divergence under mixture
Extension to F1 risk	1 week	Requires relating testing error to F1
Large-deviations refinement	2-3 weeks	Cramer-Chernoff analysis, may require new tools
State aggregation and multi-class	1 week	Extending single-state to multi-state
Writing and verification	2 weeks	Writing up the 5-10 page proof in JMLR style
Total	**7-10 weeks**	**~2-2.5 months full-time**

7.2 Optimistic Estimate (If All Technical Choices Work)

Component	Effort
Single-state testing lower bound	1 week

Component	Effort
F1 conversion	2 days
Multi-state extension	3 days
Writing	1 week
Total	**~3 weeks**

7.3 What the JMLR Framework Says vs. Reality

Claim in PAPER_FRAMEWORK	Reality
"Le Cam + Assouad minimax lower bound: 4 days"	Achievable for -regime ONLY
"Matching lower bound, constant factors"	Rate optimality is achievable; exact constants are >1 month
"Critical for JMLR acceptance"	Likely true — but the difficulty is 5-10x higher than estimated

Recommendation: Separate the two problems. The -regime minimax claim (matching Theorem 2) is essentially already done and takes 1 week to formalize. The M-regime matching lower bound (matching Theorem 1) should be presented as a separate, partially open result.

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8. Dependencies and New Tools Required

8.1 Does This Need New Assumptions?

No new assumptions needed for a rate-optimality result (exponent in M): - B1-B3 (or A1-A6) are sufficient for the lower bound construction - The lower bound holds for a SUBCLASS of distributions satisfying these assumptions → thus the worst-case over all such distributions is at least as hard - No additional restrictive assumptions needed

New assumptions that would help: - **Symmetric experts** (all have same error rate): Makes lower bound construction cleaner - **Two-class classification** (K = 2): Avoids the mixture-over-classes complication - **Known state partition**: Removes state estimation error from the minimax bound

These can be presented as the setting for the lower bound: "Even in the simplest case (symmetric experts, binary classification, known state structure), no detector can surpass exp(-2MΔ^2)."

8.2 New Mathematical Tools Required

Tool	Status	Source
Le Cam's two-point lemma	Well-known	Tsybakov (2009), Lec 2
^{2} divergence tensorization	Well-known	Polyanskiy & Wu (2022+)
KL divergence for mixtures	Standard	Van der Vaart (1998)

Tool	Status	Source
Cramer-Chernoff large deviations	Well-known	Dembo & Zeitouni (1998)
Berry-Esseen for binomial	Well-known	Not needed for the rate, only for constants
F1 risk lower bound via testing error	**Need to derive**	New (specific to this problem)
Mixture-to-product TV bound	**Need to derive**	New (mixture of product vs product)

Key technical lemma needed (not in standard textbooks):

> **Lemma (Mixture-Product TV bound).** Let P = *m Bernoulli(p) and P = (1/L)*• { =1}^L _m Bernoulli(p _). Then: `TV(P , P )  (1/2)•√( ²(P || (1/L)• _ P _ ))` where `²(P || (1/L)• _ P _ )  (1/L)• _ [((p -p _ )²/(p _ (1-p _ )) + 1)^M - 1]`

This lemma would be the technical core of the proof and requires careful verification.

8.3 Software/Tools

- None needed (pure mathematics)
- Auxiliary: could use Mathematica/SymPy to verify the ² divergence calculations for specific parameter values

9. Risk of Being Scooped

9.1 Related Literature

Paper	Topic	Overlap with SCX
Gao et al. (2016, AoS): "Minimax optimal convergence rates for estimating the ground truth from multiple annotators"	Crowdsourcing minimax rates	**Moderate**. They study minimax rates for estimating TRUE LABEL from multiple noisy annotators — different from NOISE DETECTION. Their setup is purely categorical (no features), unlike SCX's state-conditioned approach.
Berend & Kontorovich (2015, COLT): "Consistency of weighted majority votes"	Weighted majority vote convergence	**Low**. They study convergence of majority vote to ground truth.
Jaffe et al. (2015, NIPS): "Instance-dependent label-noise"	Instance-dependent label noise rates	**Low**. Theoretical bounds on learning with instance-dependent noise.

Paper	Topic	Overlap with SCX
Bhatt et al. (2017, JMLR): "Minimax-optimal label noise"	Minimax rates for learning with label noise	**Low distances**. They study learning a CLASSIFIER under label noise (Menon et al.'s setting), not noise detection.
Karger et al. (2011, STOC): "Minimax rates for crowdsourcing"	Minimax rates for Dawid-Skene model	**Moderate**. They study optimal rate for recovering ground truth labels from multiple annotators with confusion matrices.

9.2 Assessment

Risk Factor	Assessment
Direct competition	**Very low**. No one is studying the problem of "detecting which samples have label noise using multi-expert consistency" from a minimax perspective.

Risk Factor	Assessment
Adjacent competition	**Low**. The crowdsourcing minimax literature (Gao, Karger, etc.) studies a different estimand (ground truth label recovery, not noise detection).
Methodological competition	**Medium**. A statistical theorist could independently derive minimax rates for "testing whether a given sample has label noise using M independent classifiers." This is a natural problem that any PhD student in theoretical ML could encounter.
Time risk	**Increasing**. As ensemble methods and data quality become more prominent, someone may formalize this. The SCX framework's specific structure (state conditioning, F1 analysis) is unique, but the core minimax rate for testing binomial proportions with M experts is not.

9.3 Mitigation

1. **arXiv Paper 1 first**: The Theorem 1 upper bound establishes priority for the problem formulation
2. **Defensible uniqueness**: The state-conditioned structure (the Π par-

tition, A5-A6) is unique to SCX and unlikely to be independently discovered

3. **Don't delay too long**: A matching lower bound is a nice-to-have but not blocking for submission. If the 3-week proof attempt hits the 4-month mark, submit without it and note it as an open problem.

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10. Recommended Path Forward

Phase 1 (Week 1-2): Quick Wins

- Formalize the -regime minimax claim (matching Theorem 2) — this is almost done
- Publish the "large-deviations rate optimality" claim as a conjecture in the JMLR paper
- These require minimal effort

Phase 2 (Week 3-6): Core M-Regime Lower Bound

- Prove the Mixture-Product TV bound lemma (Section 8.2)
- Apply Le Cam for the single-state, two-class, symmetric-expert case
- Extend to multi-class with A6
- This yields a rate-optimality result: `liminf(-1/M) log(lower bound) = 2Δ²`

Phase 3 (Week 7-10): F1 Extension

- Convert testing error to F1 via `1 - F1` • `(FN rate)`

- Tighten constants if possible
- Write up in JMLR-proof style

Phase 4 (Optional, Beyond Paper 3): Constant Refinement

- Berry-Esseen analysis for finite-M constants
- Chernoff exponent exact constant
- Possibly a separate paper in a statistics journal

Contingency: If Stuck

If the mixture-product divergence calculation proves intractable after 6 weeks, fall back to:

1. **Weaker claim**: "SCX achieves the optimal RATE (exponential convergence in M), though the exact constant may be improvable"
2. **Conjecture**: "We conjecture that the exponent $2M\Delta^2$ is optimal based on large-deviations theory (Dembo & Zeitouni, 1998)"
3. **Empirical evidence**: "Simulations show the F1 lower bound tracks the empirical F1 within a factor of 2 for M 20"

This is acceptable for JMLR/TMLR. Not every paper needs matching lower bounds for every theorem.

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11. Summary Table

Dimension	Rating	Note
Feasibility (rate optimality)	MEDIUM	Provable with 7-10 weeks work
Feasibility (exact constants)	LOW	May require 3 months, Berry-Esseen
Feasibility (-regime)	HIGH	Already essentially done (Thm 2)
Novelty to JMLR	HIGH	Would distinguish the paper
Difficulty	7/10	Mixture-product divergence is the crux
Reviewer expectation	MODERATE	Some will ask; most will accept a conjecture
Scoop risk	LOW	But increasing over time
Effort estimate	7-10 weeks full-time	1 trained mathematical statistician
Blocking for submission	NO	Can submit without it